

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

Hardware Issue	Typical Symptom
1. No Display on Monitor	Blank screen, no POST beep, monitor shows "No Signal".
2. Faulty RAM	Computer does not boot, continuous beep sounds, or random system crashes.
3. Hard Disk/SSD Failure	Operating system does not load, boot errors, or files cannot be accessed.
4. SMPS (Power Supply) Failure	PC does not turn on, no power LED, and no fan movement.
5. CPU Overheating	Computer shuts down unexpectedly, restarts automatically, or runs very slowly.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Step	Troubleshooting Plan	What to Check
1	Check the power outlet.	Ensure the wall socket is supplying power by testing it with another device.
2	Inspect the charger and power cable.	Look for cuts, loose wires, bent connectors, or any visible damage.
3	Check the laptop charging port.	Inspect the charging port for dust, dirt, or a loose/broken connector.
4	Test the charger with a multimeter.	Measure the output voltage to verify it matches the charger's rated voltage.
5	Check the battery connection.	If the battery is removable, remove and reinsert it securely.
6	Try another compatible charger.	Use a working charger to determine if the original charger is faulty.
7	Observe the charging indicator.	Check whether the charging LED turns on when the charger is connected. If it does not, the charger, battery, or charging circuit may be faulty.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

Item	Observation
Device Tested	Standard USB Phone Charger
Measured Voltage	5.02 V DC
Expected Voltage Range	4.75 V – 5.25 V DC
Result	5.02 V is within the expected range. The USB charger is working normally.

4.A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Testing the 12V Rail of an ATX Power Supply Using a Multimeter

Step	Description
1.	Turn off the PC and disconnect the ATX power connector from the motherboard.
2.	Set the multimeter to DC Voltage (20V range) .
3.	Insert the black probe into any black (Ground) wire of the ATX connector.
4.	Touch the red probe to the yellow wire (+12V) of the ATX connector.
5.	Read and record the voltage displayed on the multimeter.

Measured Voltage

12.08 V DC

Expected Voltage Range

11.5 V – 12.6 V DC

Result

The measured voltage is **12.08 V**, which is **within the normal range**. This indicates that the **12V rail of the power supply is working properly**.

If the Voltage is Below 11.5V

If the measured voltage is **below 11.5 V**, it may indicate a **faulty or failing SMPS (Power Supply Unit)**. This can cause **random PC shutdowns, system instability, unexpected restarts, or failure to boot** because the power supply cannot provide stable voltage to the hardware.